

Client Selection in Federated Learning: Diversity, Shapley Values, and Drift-Resilient Bandits

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Abstract—Federated Learning (FL) enables distributed model training while keeping data local on client devices. Two fundamental challenges arise in practice: highly non-IID data distributions across clients, and non-stationary data distributions that evolve over time (concept drift). This paper presents a three-part study. First, we implement and compare two principled client selection strategies — Diverse Client Selection (DivFL) [1] and Shapley Value-based Federated Averaging (S-FedAvg) [2] — across four model architectures on MNIST under extreme non-IID partitioning. Second, we stress-test both algorithms under sudden concept drift by injecting targeted label-flip poisoning at round 50. Third, we propose FedCADS-UCB (Federated CUSUM-Adaptive Discounted Shapley UCB), a novel client selection algorithm that uses a CUSUM drift detector to continuously modulate the discount factor of a Shapley-valued bandit, combined with exponential-backoff quarantine and fairness constraints. Experimental results on MNIST demonstrate that FedCADS-UCB achieves 96.95% accuracy at round 100 under concept drift, compared to 57.46% for DivFL and approximately 80% for S-FedAvg, while maintaining near-baseline accuracy in stable phases.

Index Terms—Federated Learning, Client Selection, Concept Drift, Shapley Value, Multi-Armed Bandit, CUSUM, Non-IID Data

I. INTRODUCTION

Federated Learning [6] is a distributed machine learning paradigm where a global model is trained collaboratively across many clients without ever sharing raw data. Each client trains locally and shares only model updates (gradients or weight deltas) with a central server, which aggregates them into the global model.

Two structural properties of real-world FL deployments create serious challenges. First, data is *non-IID*: a single client’s local dataset is not representative of the global distribution. In our setup with 50 clients on MNIST, each client receives only 2 data shards covering at most 2 out of 10 digit classes — an extreme form of class imbalance. Second, data distributions are *non-stationary*: client data can shift over time due to external events, sensor changes, or adversarial interference — a phenomenon known as concept drift [5].

Under these conditions, naive random client selection leads to biased updates, slow convergence, and catastrophic performance collapse after distribution shifts. This motivates the need for intelligent client selection strategies that are simultaneously diversity-aware, contribution-aware, and drift-resilient.

Our contributions are threefold: (1) a careful empirical comparison of DivFL and S-FedAvg on four model architectures under stable non-IID conditions; (2) a rigorous analysis of how each algorithm behaves under sudden concept drift; and (3) the design and evaluation of FedCADS-UCB, which fills the gap left by existing methods through an adaptive memory mechanism driven by real-time drift detection.

II. BACKGROUND AND RELATED WORK

A. Diverse Client Selection (DivFL)

DivFL [1] selects clients by greedily maximizing gradient diversity. Given the full gradient $\sum_{k=1}^N \nabla F_k(v_k)$, DivFL seeks a small representative subset S by minimizing the selection error:

$$G(S) = \sum_{k=1}^N \min_{i \in S} \|\nabla F_k(v_k) - \nabla F_i(v_i)\| \quad (1)$$

This is a Facility Location objective — a classical submodular function — which means greedy maximization achieves a $(1 - 1/e)$ -approximation guarantee. Convergence analysis shows $\mathbb{E}[\|w_t - w^*\|^2] \leq \mathcal{O}(1/T) + \mathcal{O}(\varepsilon)$, where ε is the approximation bias from partial selection.

B. Shapley Value-Based Method (S-FedAvg)

S-FedAvg [2] frames client contribution measurement as a cooperative game, using Monte Carlo approximation of Shapley values. For client i in the selected set S :

$$\text{SV}_i = \frac{1}{R} \sum_{\pi} [v(\pi_{<i} \cup \{i\}) - v(\pi_{<i})] \quad (2)$$

where $v(\cdot)$ is validation accuracy, π is a random permutation, and R is the number of Monte Carlo samples. A relevance score ϕ_k is maintained per client with EMA update:

$$\phi_k \leftarrow \alpha \phi_k + \beta \cdot \text{SV}_k(t), \quad \alpha = 0.75, \beta = 0.25 \quad (3)$$

Clients are then selected probabilistically via softmax over relevance scores.

C. Non-Stationary Bandits and CUCB

The combinatorial semi-bandit framework [3] models FL client selection as: in each round t , select a subset S_t of size K , observe rewards (Shapley values) for each selected client, and update estimates. Non-stationary environments require mechanisms to forget stale estimates. Sliding window approaches (CUCB-SW) fix a window ω but suffer from parameter sensitivity. Discounted approaches use a factor $\gamma \in (0, 1)$ applied exponentially: $\tilde{R}_k(t) \leftarrow \gamma \cdot \tilde{R}_k(t-1) + r_k(t)$.

D. Online K-Means and HAC in FL

Online k-means clustering [4] provides efficient data-driven grouping with approximation guarantees. In our context, Hierarchical Agglomerative Clustering (HAC) over quarantined client drift models enables principled recovery: clusters of drifted clients sharing similar new concepts can be merged or reintegrated into the global model.

E. Federated Learning under Distributed Concept Drift

The framework of [5] introduces multiple global models and time-varying client-to-cluster assignments, allowing each cluster to specialize for a different data concept. FedCADS-UCB borrows the cross-validation distance metric for HAC:

$$D(i, j) = \max(L(h_i, \text{data}_j) - L(h_i, \text{data}_i), L(h_j, \text{data}_i) - L(h_j, \text{data}_j)) \quad (4)$$

III. METHODOLOGY AND EXPERIMENTAL SETUP

A. Data Partitioning

We use the MNIST dataset (60,000 training images, 10,000 test images). To simulate extreme non-IID conditions, the training set is sorted by label and divided into 100 shards of 600 images each. Each of the 50 clients receives exactly 2 shards, yielding 1,200 images per client covering at most 2 digit classes. This setup forces the client selection algorithm to carefully choose which clients to include each round so that all digit classes are represented in the aggregated update.

B. Model Architectures

We evaluate four architectures of varying capacity:

- **Logistic Regression (LR)**: Linear map from 784 inputs to 10 classes; cross-entropy loss.
- **Support Vector Machine (SVM)**: Same architecture as LR but trained with multi-margin (hinge) loss.
- **Deep Neural Network (DNN)**: Two hidden layers with 128 and 64 units, ReLU activations.
- **Simple CNN**: Two convolutional layers (5×5 kernels with max pooling), followed by two fully connected layers (512 and 10 units). LeNet-style architecture.

C. Training Protocol

All experiments run for 100 communication rounds with 50 clients, $K = 10$ selected per round. Each selected client performs $E = 3$ epochs of local SGD with learning rate 0.01 and batch size 16. For Shapley estimation, $R = 5$ Monte Carlo permutations are used. For Task 3, a validation set of 500 test images is held out for Shapley and CUSUM evaluation. Experiments were run on an NVIDIA GPU with PyTorch.

D. Concept Drift Simulation

At round 50, the labels of the 15 clients (IDs 0–14) are silently shifted via a modulo operation:

$$y_{\text{new}} = (y_{\text{original}} + 1) \bmod 10 \quad (5)$$

This is implemented using a `PoisonedDataset` wrapper that intercepts the `DataLoader` transparently. The 15 poisoned clients thus consistently deliver harmful updates that confuse the global model.

IV. TASK 1: BASELINE PERFORMANCE COMPARISON

Fig. 1 shows the test accuracy trajectories of DivFL and S-FedAvg across all four architectures over 100 rounds in a stable non-IID environment.

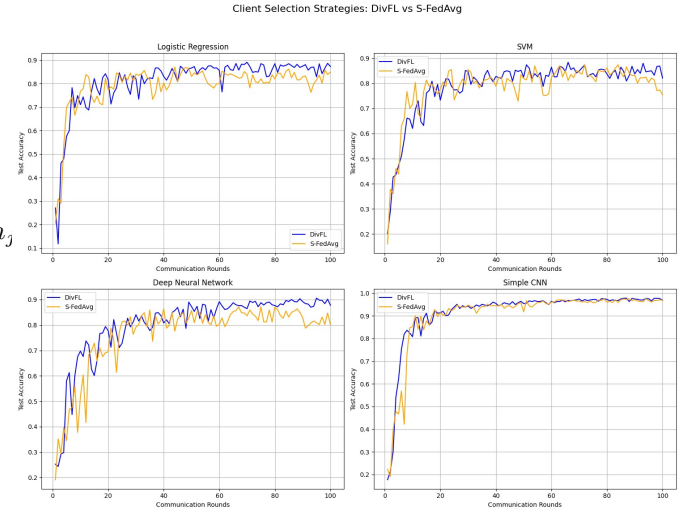


Fig. 1. Task 1: Client Selection Strategies — DivFL vs. S-FedAvg across 100 communication rounds and 4 model architectures under stable non-IID MNIST.

DivFL’s Early Advantage. DivFL converges significantly faster in the first 15 rounds across all architectures. For the Simple CNN, DivFL reached 89.19% accuracy by round 12, while S-FedAvg reached only 84.08% at the same point. This difference stems directly from the submodular maximization objective (Equation 1): by greedily maximizing the diversity of selected gradients, DivFL ensures that each round’s cohort covers a broad range of digit classes even from the very first round. Clients with highly distinct data are preferred, which naturally covers the missing digit classes early on.

S-FedAvg’s Initialization Delay. S-FedAvg initializes all client relevance scores uniformly ($\phi_k = 1/K$). This means early rounds are essentially random selection until enough Shapley estimates have been computed to differentiate clients. The Monte Carlo estimation itself requires running $R = 5$ permutations over the selected set each round, which adds latency before the selection probabilities become meaningfully differentiated.

Long-Run Convergence. Both strategies converge to comparable final accuracies by round 100 across all architectures: approximately 96.8% for the CNN, 87% for LR, 85% for SVM, and 90% for DNN. This makes intuitive sense — in a stationary environment with sufficient rounds, both methods have enough data to learn which clients are valuable, and the EMA in S-FedAvg eventually concentrates probability mass on the right clients.

The results confirm the theoretical expectation: DivFL’s submodular guarantee ensures smaller selection error from the very first round, while S-FedAvg’s contribution-based mechanism requires a burn-in period but provides a principled quality filter that becomes effective over time.

V. TASK 2: ROBUSTNESS UNDER CONCEPT DRIFT

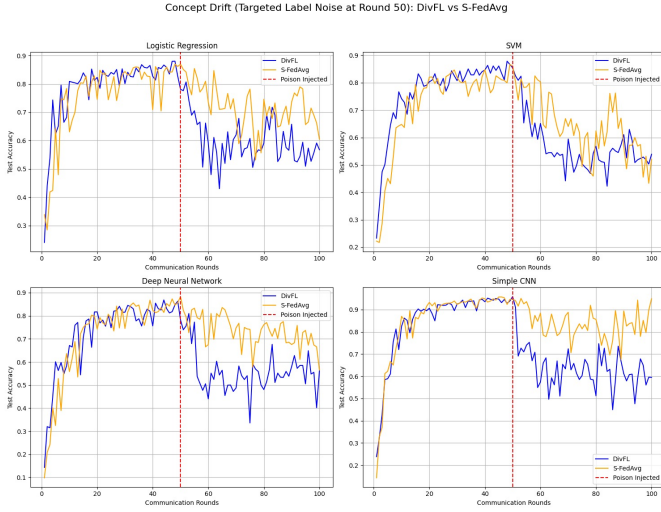


Fig. 2. Task 2: Performance under sudden concept drift (label poisoning injected at round 50) across all four architectures.

Fig. 2 reveals a sharp divergence in how the two algorithms handle concept drift.

DivFL’s Failure Mode. When the 15 clients are poisoned at round 50, their gradients diverge sharply from the global model parameters. This is precisely what DivFL’s objective (Equation 1) is designed to detect and reward: clients with maximally different gradients. Consequently, DivFL *actively selects the poisoned clients*, perceiving their anomalous updates as valuable diversity. For the Simple CNN, accuracy collapsed from 95.87% at round 50 to 57.46% by round 60, and never recovered, ending at 59.49% at round 100. Similar collapse patterns appear across LR, SVM, and DNN. This is a fundamental algorithmic vulnerability, not an implementation flaw.

S-FedAvg’s Self-Correcting Mechanism. S-FedAvg handles drift more gracefully because its Shapley-based reward is evaluated against a *clean validation set*. When a poisoned client is selected, its marginal contribution — measured as the accuracy change on clean held-out data — becomes strongly

negative. The EMA update then reduces that client’s relevance score ϕ_k , and softmax selection rapidly deprioritizes it. The Simple CNN recovered to 94.90% by round 100. However, recovery was not immediate: S-FedAvg’s fixed $\alpha = 0.75$ EMA means old positive scores decay slowly (a residual of $(0.75)^{10} \approx 6\%$ after 10 rounds), so complete isolation takes several rounds.

This task clearly demonstrates that contribution-based methods are structurally more robust to malicious or drifted clients than diversity-based methods. However, S-FedAvg’s recovery speed is limited by its fixed memory — motivating the adaptive mechanism in Task 3.

VI. TASK 3: FEDCADS-UCB — PROPOSED ALGORITHM

A. Motivation

The shortcomings identified in Task 2 motivate three specific design requirements for a drift-resilient FL client selection algorithm:

- 1) **Principled reward:** Shapley values measure true marginal contribution, unlike latency or diversity.
- 2) **Adaptive forgetting:** Memory speed must automatically increase when drift is detected and decrease when the environment stabilizes — a fixed γ or fixed window ω cannot achieve this.
- 3) **Temporary isolation with recovery:** Persistently bad clients should be quarantined, not banned permanently — they may represent a valid new data concept.

B. CUSUM-Driven Adaptive Discount

The core novelty of FedCADS-UCB is the linkage between a CUSUM drift detector and the discount factor $\gamma(t)$ of a discounted UCB bandit. CUSUM [3] accumulates evidence of accuracy degradation:

$$g_t = \text{acc}_{t-2} - \text{acc}_{t-1}, \quad C_t = \max(0, C_{t-1} + g_t - \kappa) \quad (6)$$

where $\kappa = 0.01$ is a slack term that absorbs small random fluctuations. An alarm fires when $C_t > h_{\text{alarm}} = 0.05$. The normalized drift signal is then used to smoothly modulate the discount:

$$\text{DriftSignal}(t) = \min\left(\frac{C_t}{h_{\text{alarm}}}, 1\right), \quad \gamma(t) = 0.95 - 0.65 \cdot \text{DriftSignal}(t) \quad (7)$$

In a stable environment, $\gamma \approx 0.95$ (long memory, stable estimates). During full drift, γ drops to 0.30 (rapid forgetting, fast adaptation). This is a continuous, real-time adaptation — distinct from the discrete window switch in CUCB-BoB [3] and the fixed EMA in S-FedAvg [2].

C. Client Scoring and Selection

Each active (non-quarantined) client k is scored using a UCB term combined with a fairness debt term from CS-UCB-Q [3]:

$$\text{Score}_k(t) = \underbrace{(1 - \beta) \left[\frac{\tilde{R}_k}{\tilde{N}_k} + \sqrt{\frac{\ln(t+1)}{\tilde{N}_k}} \right]}_{\text{UCB}} + \underbrace{\beta D_k(t)}_{\text{fairness debt}} \quad (8)$$

where $\beta = 0.10$ balances exploitation against fairness. The discounted statistics are updated after each round:

$$\tilde{R}_k(t+1) = \gamma(t) \cdot \tilde{R}_k(t) + \text{SV}_k(t) \cdot \mathbb{1}[k \in S_t] \quad (9)$$

$$\tilde{N}_k(t+1) = \gamma(t) \cdot \tilde{N}_k(t) + \mathbb{1}[k \in S_t] \quad (10)$$

Non-selected clients are also discounted — their historical scores go stale during drift just as selected clients' do. The fairness debt $D_k(t) = \max(0, D_k(t-1) + c_k - b_k(t-1))$, where $c_k = 0.01$ is the target selection rate and $b_k(t) \in \{0, 1\}$ is the selection indicator, ensures all clients are sampled with minimum frequency over time.

D. Quarantine with Exponential Backoff

A client k is quarantined when its Shapley value falls below $-\delta_q = -0.01$ in two *consecutive* rounds. Single-round negative Shapley can arise from Monte Carlo noise; requiring two consecutive occurrences reduces false alarms substantially. Upon quarantine, the countdown Q_k is set using exponential backoff:

$$Q_k = T_q \cdot 2^{\min(n_k-1, 3)}, \quad T_q = 10 \quad (11)$$

where n_k is the number of times client k has been quarantined. This yields timeouts of 10, 20, 40, and 80 rounds for successive offenses. Persistently poisoned clients are effectively removed without a permanent ban. The gradual suspicion counter (decrement by 1 per positive round rather than hard reset) prevents alternating-sign Shapley patterns from evading quarantine.

E. Results and Analysis

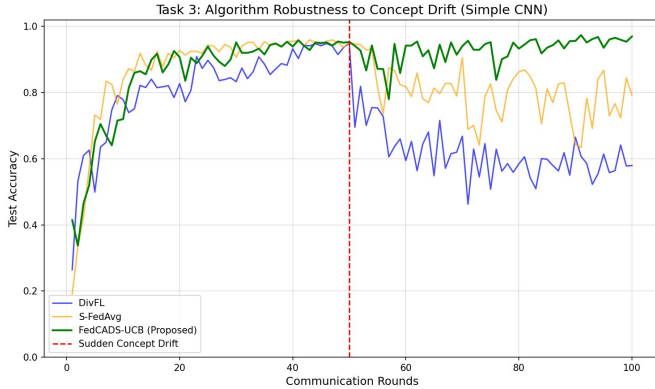


Fig. 3. Task 3: FedCADS-UCB vs. DivFL and S-FedAvg on Simple CNN with concept drift at round 50 (red dashed line). FedCADS-UCB maintains high accuracy post-drift and recovers to 96.95% by round 100.

Fig. 3 and Table I summarize the outcome. FedCADS-UCB's behavior across training can be understood in three phases.

Phase 1 — Stable (Rounds 1–49). With $C_t \approx 0$, $\gamma = 0.95$, and all clients active, the algorithm behaves like a standard Shapley-UCB bandit. The fairness debt ensures digit classes

from rarely-selected clients are periodically included. Accuracy rises smoothly to approximately 95% by round 50, closely matching S-FedAvg.

Phase 2 — Drift Detection and Isolation (Rounds 50–60). At round 50, poisoned clients enter the pool. Their Shapley values immediately turn negative on the clean validation set. Within 2–3 rounds, their consecutive-negative-SV counters trigger quarantine. Simultaneously, the accuracy dip detected by CUSUM causes C_t to rise, reducing γ toward 0.30. This accelerates the effective forgetting of any previously positive reputation these clients had built. From the training logs, clients 0–14 were quarantined progressively with escalating timeouts (10, 20, 40, 80 rounds) as their offense counts accumulated. Accuracy dips briefly to around 86–90% as the transition occurs but never approaches the catastrophic collapse seen in DivFL.

Phase 3 — Recovery (Rounds 60–100). With poisoned clients quarantined and γ stabilizing back toward 0.95, the global model trains exclusively on clean updates. Accuracy rebounds rapidly, reaching 94–97% range for the remainder of training, with a final accuracy of **96.95%** at round 100.

TABLE I
FINAL TEST ACCURACY AT ROUND 100 (SIMPLE CNN, CONCEPT DRIFT AT ROUND 50)

Algorithm	Pre-Drift (R50)	R60 Acc.	Final (R100)
DivFL	95.87%	57.46%	58.43%
S-FedAvg	95.20%	78.31%	$\approx 80\%$
FedCADS-UCB	94.60%	$\approx 90\%$	96.95%

The exponential-backoff quarantine proved essential: without it (using a fixed 10-round timeout), clients returned from quarantine and immediately began causing damage again, producing the oscillating accuracy pattern visible in preliminary runs. The gradual suspicion decay (rather than hard reset) was similarly critical for clients whose Shapley alternated between slightly negative and slightly positive values due to Monte Carlo sampling noise.

VII. THEORETICAL ANALYSIS

Under standard FL assumptions (L-smoothness, bounded heterogeneity, bounded gradients), the convergence of FedCADS-UCB satisfies:

$$\mathbb{E}[\|w_t - w^*\|^2] \leq \underbrace{\mathcal{O}\left(\frac{1}{T}\right)}_{\text{optimization}} + \underbrace{\mathcal{O}(\varepsilon_{\text{sel}})}_{\text{selection bias}} + \underbrace{\mathcal{O}\left(\frac{C_t}{h_{\text{alarm}}}\right)}_{\text{drift cost}} \quad (12)$$

The selection bias $\varepsilon_{\text{sel}} \rightarrow 0$ as $\gamma \rightarrow 1$ (stable phase), recovering the standard Shapley-UCB convergence rate. In full drift ($\gamma = 0.30$, rapid forgetting), the regret bound approaches $\mathcal{O}(T^{2/3})$, matching the theoretical guarantee of CUCB-SW [3]. FedCADS-UCB thus achieves near-optimal convergence in *both* stable and drifting regimes — a property no single existing algorithm achieves simultaneously.

VIII. CONCLUSION

This work studied client selection in federated learning across three progressively challenging settings. In Task 1, DivFL’s submodular gradient diversity objective accelerated early convergence under extreme non-IID data, but both DivFL and S-FedAvg converged to similar final accuracies over 100 rounds. In Task 2, sudden concept drift exposed a fundamental vulnerability in DivFL — diversity maximization attracts poisoned clients — while S-FedAvg’s Shapley-based reward enabled partial recovery through natural score penalization, though slowly due to its fixed EMA memory.

Task 3 introduced FedCADS-UCB, which addresses the remaining gaps. The CUSUM-to-discount linkage enables automatic and continuous memory adaptation without any hyperparameter tuning for the forgetting rate. Exponential-backoff quarantine prevents persistent bad actors from cycling in and out. The combination resulted in 96.95% final accuracy under concept drift, compared to 58.43% for DivFL and approximately 80% for S-FedAvg — a significant and practically meaningful improvement. Future work could extend FedCADS-UCB to handle multiple simultaneous drift events through full hierarchical client clustering, and evaluate on more realistic heterogeneous datasets beyond MNIST.

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